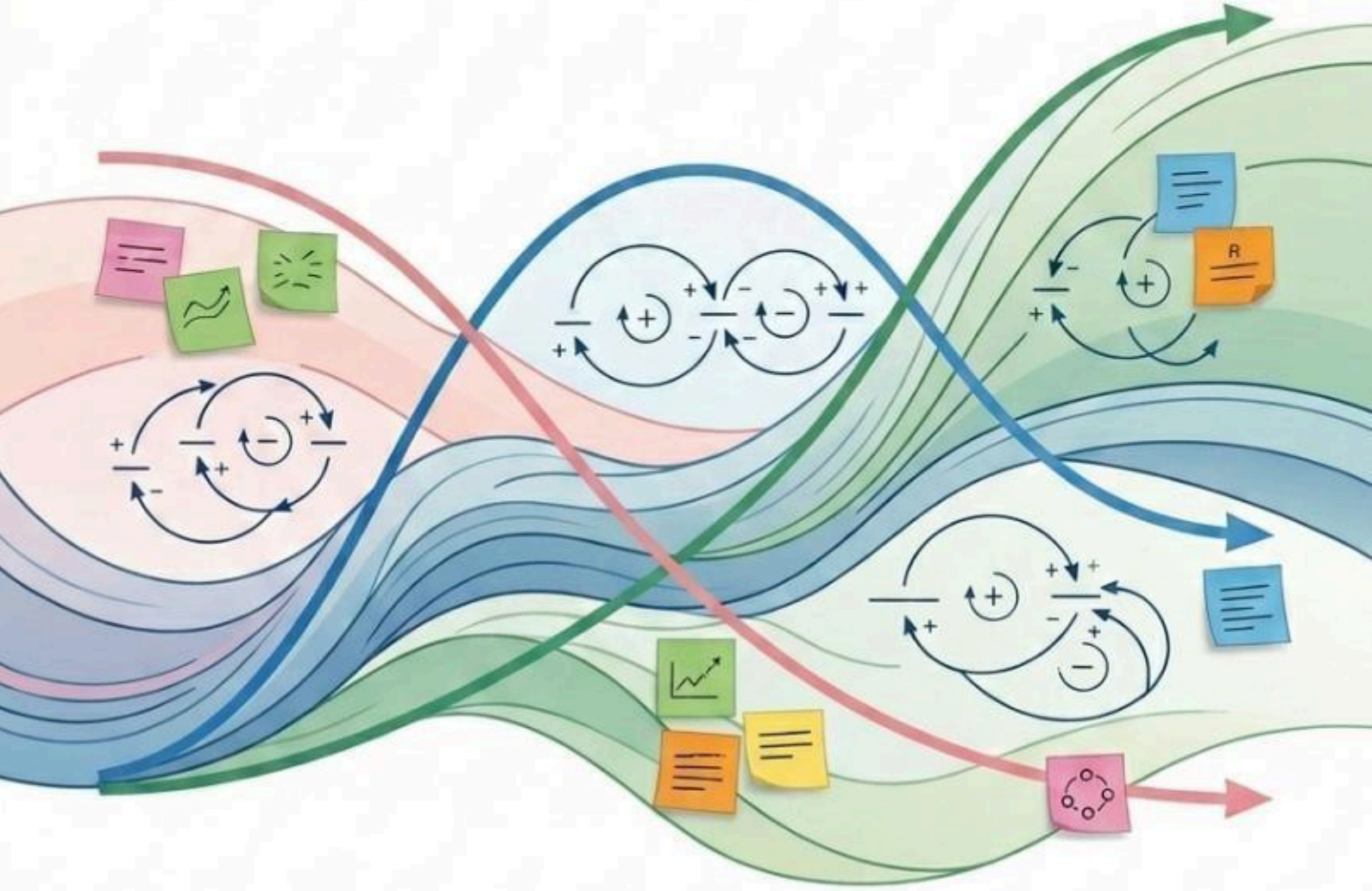
Pathways to desired futures

A guide to co-creating action plans and scenarios using the 3H-CLD approach.



Three Horizons with Causal Loop Diagrams

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INTRODUCTION

Guide objective

The objective of this guide is to teach how to organize, facilitate, and analyze data from a dialogue process based on the 3H-CLD method (Three Horizons combined with Causal Loop Diagrams), developed by researchers from the Earth System Science Division (DICST) of INPE and the Stockholm Resilience Center of Stockholm University (Colste. et al., 2023; Aguiar et al., 2025). This booklet was created in response to requests from stakeholders across sectors, following the positive results achieved in the workshops. Therefore, we sought to combine simple language with comprehensive information, without targeting a specific audience.

What is 3H-CLD?

3H-CLD is a method designed to support participatory transformation processes toward desired futures, which evolved through participatory activities led by INPE in recent decades. Although initially conceived to support discussions about sustainable and just futures at multiple scales, it can also be applied in other contexts to collectively discuss how to achieve desired futures.

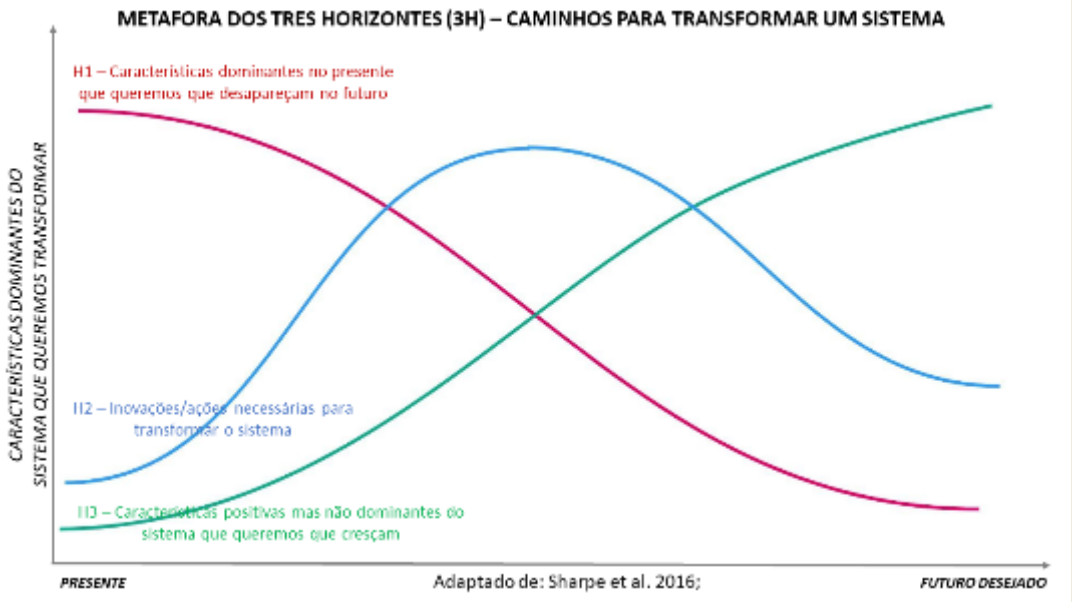
The method is a methodological evolution of future studies and systems thinking, integrating the intuitive structure of the Three Horizons (Box 1), creative processes, and the analytical rigor of systems thinking. The artistic component is essential to promote trust within the group and create a sense of ownership of the desired futures. The actions to achieve these futures are, in turn, co-constructed from an understanding of the root causes of current problems. *The basic premise of the method is that only from this systemic understanding can we identify consistent, contextualized actions with transformative potential.* This is the role of the Causal Loop Diagrams (Box 2), used so that participants, during the workshops, jointly identify cause-and-effect relationships, feedback loops, systemic structures, and leverage points. From this evolution and integration emerged the 3H-CLD (Aguiar et al., 2025).

Box 1 - The Three Horizons Diagram

The Three Horizons diagram, initially proposed by Sharpe et al. (2016), is an intuitive way to visualize and discuss transformative change. The three horizons represent, respectively:

- Horizon 1 (H1): Dominant characteristics currently in the system, which we want to disappear or lose importance in the future;
- Horizon 2 (H2): Innovations/actions needed to transform the system; and
- Horizon 3 (H3): Positive characteristics that are currently nonexistent or not dominant, and that we want to become dominant in the future;

All horizons coexist over time, and the diagram is a metaphor that helps people think about "paths" to the future, recognizing the profound uncertainty and the fact that multiple paths can coexist, interact, and diverge. The method also discusses which actors influence the necessary actions, as well as the role of power relations in transforming the system.



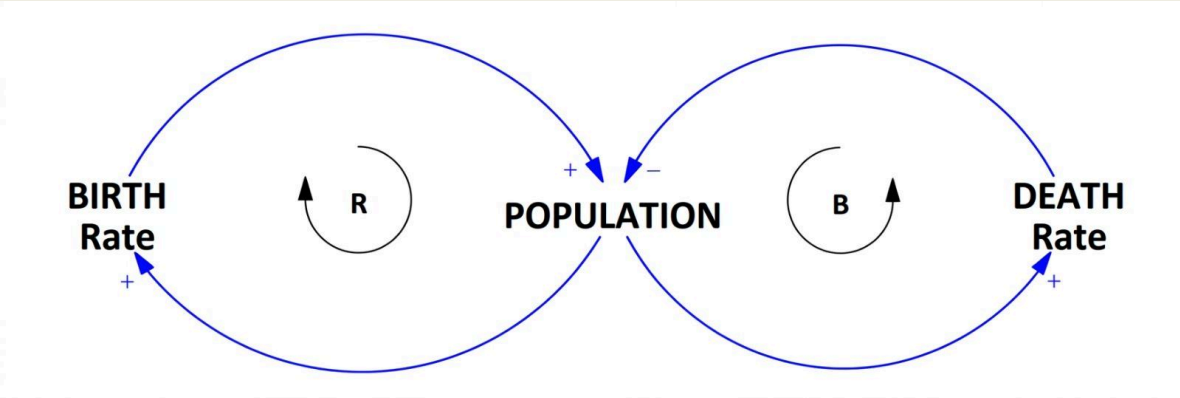
The Three Horizons diagram. In the 3H diagram, three lines are drawn on two axes. The pink line represents the current system (H1), the blue line the transition process (H2), and the green line potential future alternatives (H3). The X-axis represents the evolution over time, from the present to the future. The Y-axis represents the degree of dominance of certain characteristics, initiatives, or events of the system (Sharpe et al. 2016).

Box 2 - What is a Causal Loop Diagram (CLD)?

A Causal Loop Diagram (CLD) is a visual tool of systems thinking that helps represent how the elements of a system influence each other. Instead of looking at problems in isolation, the CLD allows you to see the interconnections, feedback loops, and structures that keep the system functioning in a certain way. In a CLD, each relevant element of the system is represented as a variable—a noun or short phrase that can "increase" or "decrease" (eg, "Population," "Pollution," "Institutional Trust"). The arrows between the variables represent causal links and carry a polarity sign. The "+" sign indicates that two variables move in the same direction. If A increases, B also increases; if A decreases, B also decreases. The "-" sign: the variables move in opposite directions. If A increases, B decreases; if A decreases, B increases. When we follow the arrows from a variable and return to it, we close a cycle—a loop.

Practical example — population dynamics

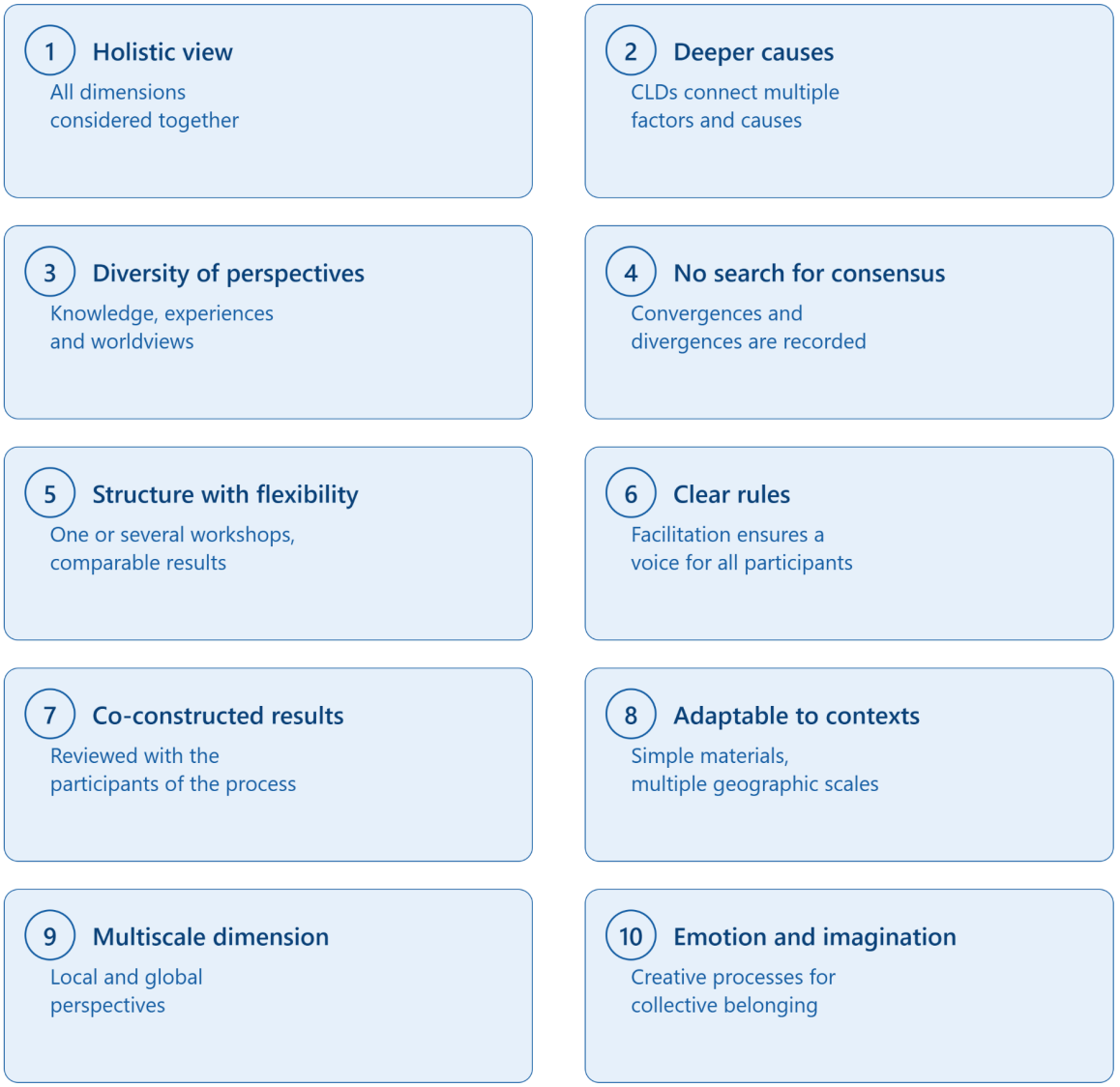
Consider the variables Population, Births, and Deaths. A larger population generates more births (+), and more births increase the population (+) — since both links are positive, a reinforcing loop (R) is closed: the larger the population, the more it grows. At the same time, a larger population also generates more deaths (+), but more deaths reduce the population (-) — there is a negative sign in the path, which characterizes a balancing loop (B): growth tends to be contained by the increase in deaths. The population's behavior over time depends on which of these two loops predominates at each moment.



Reinforcing loop (R) between Population and Births, and balancing loop (B) between Population and Deaths. The reinforcing loop amplifies changes in the same direction. It can generate accelerated growth or progressive collapse. A balancing loop (B) tends to stabilize the system by counterbalancing changes.

In the context of the 3H-CLD method, Causal Loop Diagrams are used to deepen the understanding of the root causes of present-day problems.

What are the main characteristics of 3H-CLD?

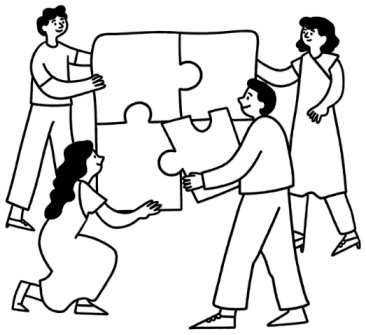


1. All dimensions of sustainability are considered holistically.
2. Problems are not analyzed in isolation. CLDs help visualize how different factors are connected, reveal deeper causes, and identify transformative actions that can influence multiple problems.
3. The process values different perspectives, knowledge, experiences, and worldviews.
4. It does not seek consensus, but captures multiple perspectives, recognizing convergences and divergences, which are noted and analyzed.

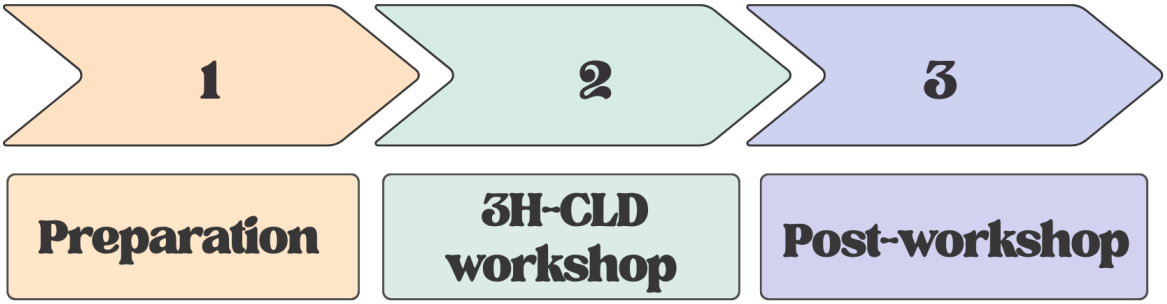
5. The method offers both structure and flexibility. A dialogue process based on the 3H-CLD method can be organized in a single workshop or in multiple workshops, with similar, easily integrable, and comparable results.
6. Clear structure and rules ensure that everyone has a voice through effective mediation.
7. The results are constructed and reviewed with the participants, seeking to preserve their perspectives and avoid the researchers' interpretations being imposed on the process (Aguiar et al., 2025; Sonetti -González et al., 2025).
8. The process is easily adaptable to multiple contexts and geographic scales, using simple materials.
9. The method offers a multiscale dimension, based on Folhes et al. (2010), and also integrates a discussion of global perspectives on pathways to sustainability, without imposition.
10. Finally, it uses emotion and imagination through creative processes to promote ownership and belonging to the desired collective future.

How is this guide organized?

This guide is organized into three workflow phases for structuring a dialogue process based on the 3H-CLD method: (1) Workshop preparation, in particular actor mapping; (2) conducting the workshops themselves; (3) post-workshop. To clarify the process and the interdependence between phases, we present the workflow below.



3H-CLD
Workflow phases



PREPARING THE WORKSHOPS

Defining the objective: Which system do we want to transform?

The method was designed to discuss sustainable and just futures in a region or system, but it can be adapted to different objectives. The workshop's focus should be included in the title of the Three Horizons Diagram, which will be visible during the workshop.

Examples: In the Bridging Values Brazilian case study, a single workshop was held with traditional communities to discuss sustainable and just futures for conservation, focusing on the Boqueirão da Onça Mosaic of protected areas (https://www.ccst.inpe.br/bridgingvalues/). In the NEXUS- XPaths project, nine workshops were held in the Brazilian case study (https://nexus.ccst.inpe.br/). Each workshop discussed the future of a specific scale and area of study within the water-energy-food nexus. The results of the multiple workshops were compared and subsequently integrated. In the AmazonFACE project (https://amazonface.unicamp.br/), however, the discussion focused not on geography, but on the future of science in the Amazon and the role of traditional knowledge, all within a single workshop.

Choosing participants: Whose desired future?

Besides the topic, another crucial decision is **which voices will be present.** As with any participatory process, this is the most critical stage.

Workshops are generally organized with 20-25 actors, divided into two or three groups of 8-12 participants each (in the case of in-person workshops[1]). In some cases, only one type of actor is involved (for example, only representatives of traditional communities). Multi-actor workshops, by contrast, seek to hear multiple perspectives, including actors from various sectors such as government, the productive sector, civil society, and academia. Some situations require considerable

[1] The method works much better in person. However, during COVID-19, workshops were conducted satisfactorily in virtual mode (Aguiar et al., 2023).

attention to actor selection and criteria for dividing groups, particularly if the objective is a multi-actor process in conflict contexts. Situations may arise that weaken the process during the workshop [2]or even make its realization unfeasible due to a lack of quorum if there is no adequate preparation.

In either case (multi-actor or focused on one type of actor), it is necessary to have a good understanding of the reality and the people, to gain their trust, and to assure participants that everyone will have a voice. In our experience, two to three months of local work to organize a 3H-CLD workshop is ideal for establishing this trust. There are several ways to select participants. We have used an actor mapping technique called NETMAP (Box 3) [3].

Box 3 - Actor mapping using the NETMAP method



One technique we have frequently used to select participants for the 3H-CLD workshops is the NETMAP method (Schiffer and Peakes, 2009; Bosch et al., 2024; Olofsson et al. , 2025). The selection process occurs in two phases. Initially, several NETMAP workshops are organized with actors from different groups or locations contacted during fieldwork. These meetings generally involve 2-5 people from the same sector, for example: (a) state and local governments; (b) companies, industries, and large landowners; (c) civil society/social movements; (d) gym. The initial selection of these actors for the NETMAP workshops should align with the objective of the process (eg, the context of conflicts in the water-energy-food nexus in a given territory).

In the NETMAP workshops, through a series of guiding questions, participants identify the main actors influencing the territory, the types of relationships between them (positive,

[2]In our experience with the NEXUS project, in situations of extreme conflict it is advisable to divide actors with very divergent perspectives into distinct groups, for reasons of comfort, safety and fluidity of discussions (Aguiar et al. , 2023).

[3]Given the impossibility of extensive fieldwork, we have the example of the methodology used by the AmazonFACE program , where contact through projects within the program and/or partner projects was fundamental for the selection and building of trust with the participants.

negative, close, or distant), and the flows of power, support, or conflict. This collective process allows participants to visualize how the local system is structured and who exerts the greatest influence on decisions that affect community life.

Examples of guiding questions during the NEXUS project workshop: (A) **Actors**: Which institutions can influence or change the system? (B) **Influence**: What influence do these actors exert/have on the system? Give 1 to 3 "dots"; (C) **Relationships**: What is the intensity of these relationships? Is it possible to define the type of relationship (hierarchical, collaborative, or information exchange)?

We used simple materials, such as cardboard or any other paper, and thick colored pens so that participants could write the names of the main actors operating in the region, identify their connections, and their degree of influence. The degree of influence was identified by assigning labels (dots); the more dots assigned, the greater the influence.

Based on the exercises, key organizations and people emerged in each context. Some names recurred across different meetings. In the NEXUS-XPATHS project, the objective was to select approximately 25 actors from different sectors for the 3H-CLD workshop. The NETMAPs were transcribed into network analysis software. The selection logic was based on the degree of centrality, as described below. Degree centrality is a simple measure we can use to gauge power within a network or social structure.

Organizing the workshop(s): location, materials, training facilitators.

In addition to the three months of work by at least one team member in mapping and selecting actors (travel, accommodation, car rental), the cost of the workshops depends on the chosen location(s) for the event(s), the need for meals, and the payment of travel and accommodation expenses for participants and facilitators. An in-person workshop with 20 actors, divided into two groups, ideally requires 4 facilitators: one facilitator and one co-facilitator per group. The method was designed to use inexpensive materials so that it can be applied in remote communities or in hotel convention rooms.

To ensure all stages are completed within the estimated time*, prepare the rooms before the sessions begin and properly train the facilitators*.



About the rooms:

- Ideally, each group should be in a separate room. In our experience, the noise from the other group is *always* disruptive.
- Preparing the rooms for each group before the session involves:
    - Drawing the 3H diagram using felt-tip pens on a 1-meter by 2-meter sheet of brown paper. Affixing it to a large table or the wall. Placing the group name and the system to be transformed at the top.

- Setting out ballpoint pens and sticky notes for participants (at least two of each color per STEP). Leaving the stacks of sticky notes on the table for distribution.
- Setting up a flip chart next to the table for noting points of divergence.
- Preparing a 1m x 1m sheet of white paper for CLDs, with the group name at the top.



Role of facilitators:

- When there is a facilitator and a co-facilitator per group, their roles must be clearly defined. The facilitator will lead the process, guide the activities, and create conditions so everyone has the opportunity to contribute. The co-facilitator will monitor the time allotted for each stage, record the discussions, including any divergences, and offer general support to the group and the facilitator throughout the process.
- **A thorough on-site rehearsal** is essential.
- **Facilitators should not express their opinions,** but only give participants the floor according to a well-defined script, detailed in the following sections.

Conducting the 3H-CLD Workshops

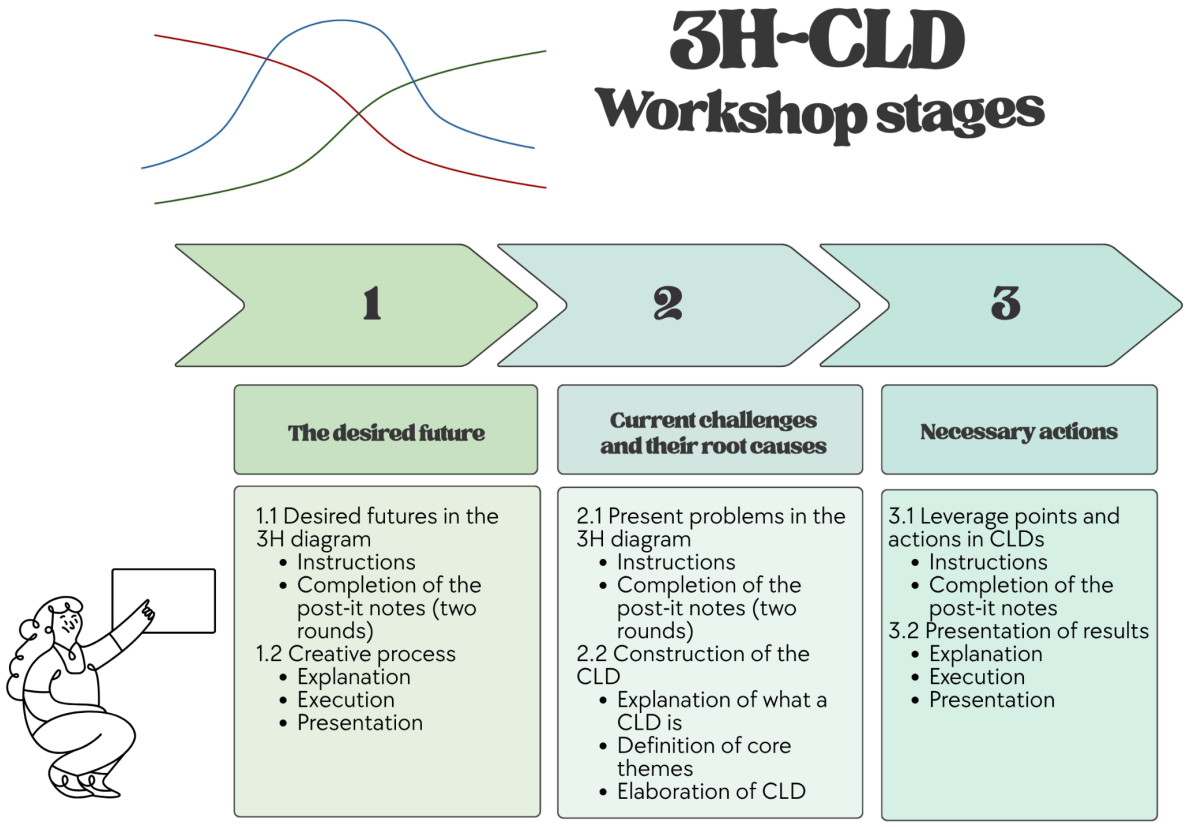
Overview

The 3H diagram (Box 1) structures the workshop activities into three stages:

STAGE 1 (focusing on H3, "The desired future")

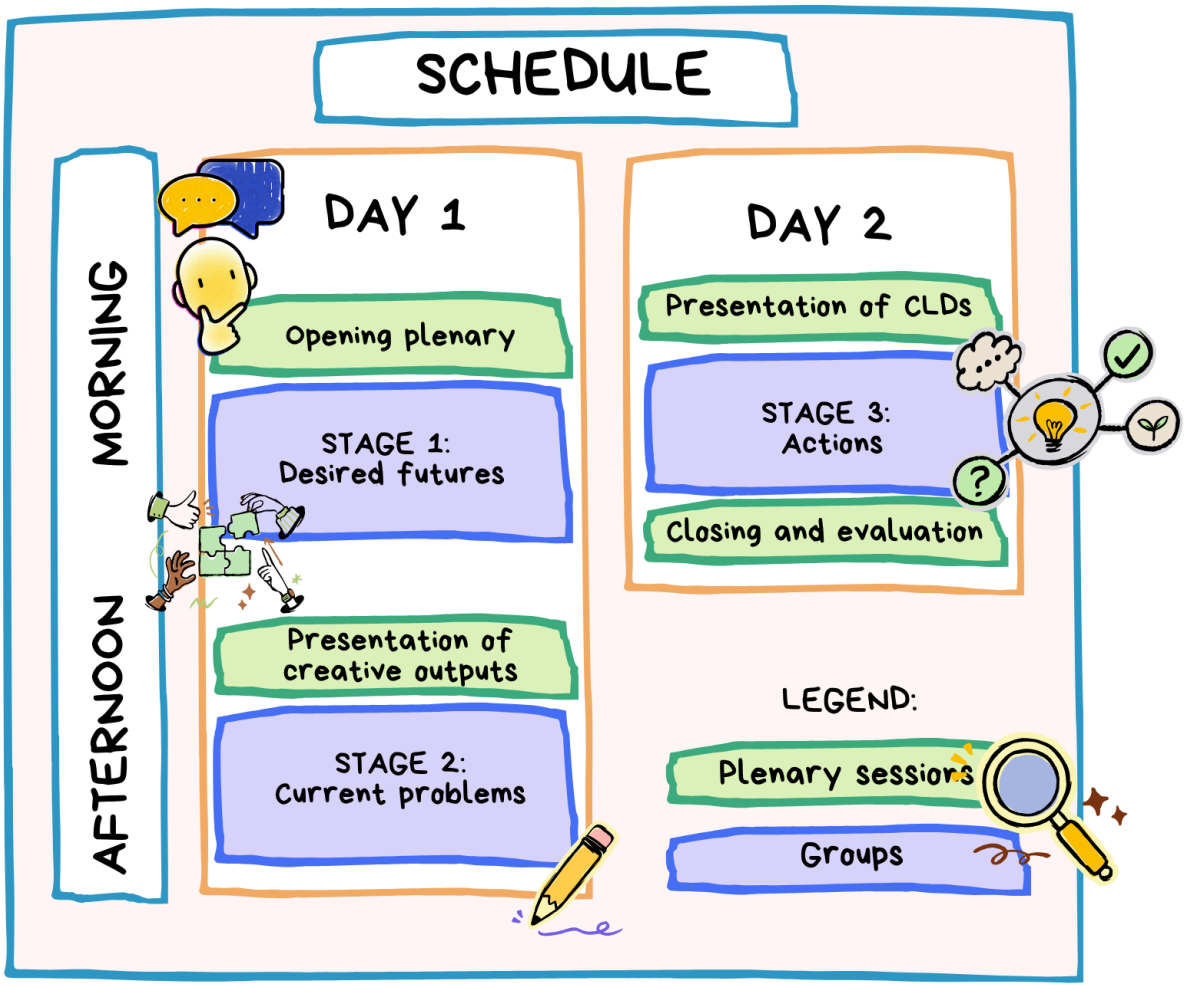
STAGE 2 (focusing on H1, "Current challenges and their root causes")

STAGE 3 (focusing on H2, "How to achieve the desired future from the present").



In STAGE 1, participants discuss their aspirations for a sustainable and just future for their region, considering all dimensions of sustainability (economic, social, environmental, governance). These aspirations are synthesized into a creative product (letter, drawing, play, etc.) that, in addition to synthesis, aims at the appropriation and internalization of the envisioned future. In STAGE 2, participants discuss current problems and their systemic causes, deepening their

understanding through systemic thinking tools such as Causal Loop Diagrams. Based on their understanding from the previous stages, in STAGE 3 they discuss integrated actions that address these causes and enable the achievement of several objectives, avoiding sectoral solutions and/or solutions that target only some of the objectives [4].



In the following subsections, each STEP is detailed. The suggested duration for each STEP is based on an in-person workshop held over a day and a half (ideally) ,

[4]In the context of sustainability, an important part of the process involves presenting, during workshops, a synthesis of global perspectives on pathways to achieve sustainability goals, fostering discussion about divergences (Aguiar et al. , 2020) . To avoid imposing or influencing the actors, these narratives are presented in the workshops only after STAGES 1 and 2, which generates discussions about divergences and convergences between global and local proposals and also supports STAGE 3. However, this stage is often not carried out in more local applications (Aguiar et al. , 2023).

although other configurations are possible [5]. To ensure each step is completed within the suggested timeframe — particularly to allow all participants to contribute equally—it is recommended to divide participants into groups of 8 to 10 people (no more than 12) and use two facilitators. ***Repeat the process described below in each group.*** The results are then integrated and compared (Post-workshop phase).

[5]Recently, the AmazonFACE project held a workshop covering all stages in just one day, due to logistical reasons. In the Bridging Values Projects, some countries have decided to divide the stages into workshops spaced a few months apart.

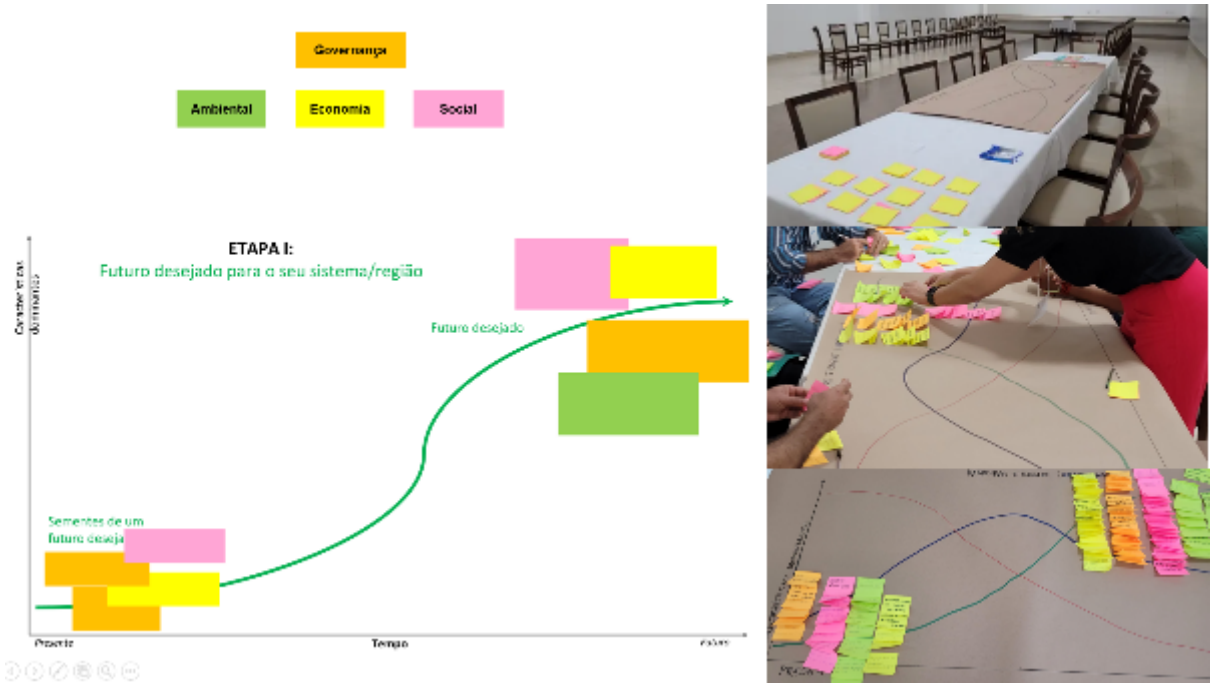


Source: Aguiar et al. 2025

Step 1: Desired Futures

Step 1.1 - Desired futures in the 3H diagram (60 min)

In each group, participants are invited to discuss the system's desired future through questions across several dimensions relevant to the chosen theme. Here, we will exemplify this with the dimensions we use in exercises focused on sustainable and just futures: social, environmental, economic, and governance. To ensure all dimensions of sustainability are addressed, distribute different-colored sticky notes at the beginning of the session (preferably already placed on the tables for easier access).



a) Instructions and initial completion of the sticky notes (15 min)

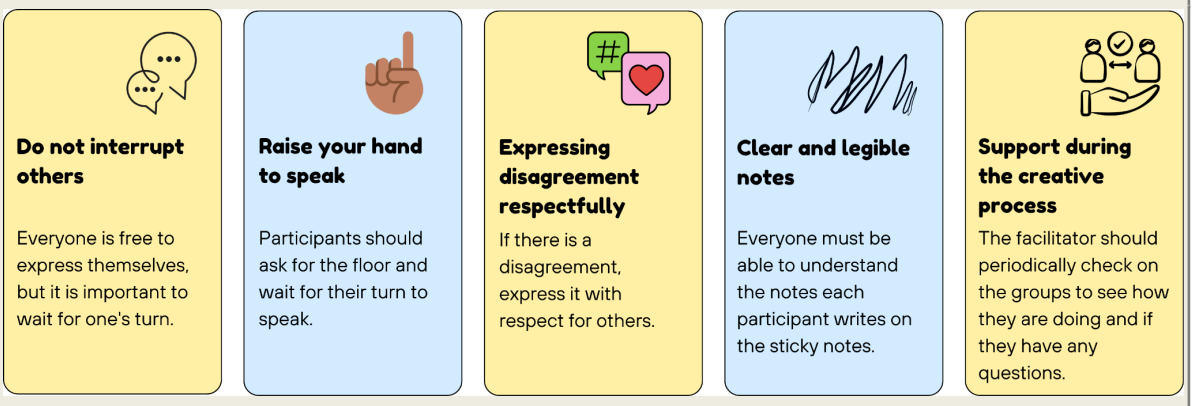
- Facilitator 1 explains the rules of the game:
    - He explains that we will start with what each person dreams of as their desired future within a defined timeframe (eg, 2050) and encourages them to dream and use their imagination.
    - Indicate on the 3H diagram, in the table, the location where we will work: upper right corner (H3), green line (Figure 1).

- Review the agreed-upon colors for the stick notes, for example, representing four dimensions: social (pink), environmental (green), economic (yellow), and governance (orange).
- It explains that consensus is not necessarily sought; it indicates the flipchart *of* potential divergences.
- It states that participants should write as individuals, and not as representatives of the organization they work for.

- Facilitator 1 asks participants to write on the sticky notes :
  - Ask them to fill out two different colored sticky notes with keywords or short phrases about their desired future, addressing the 4 dimensions.
  - Please pay attention to the handwriting, preferably in a style that is easier to read.
  - Keep the participants silent for 5 minutes while they fill out the sticky notes (they usually ask for a few more minutes).
  - Meanwhile, facilitator 2 ensures everyone has *sticky notes* and pens and assists participants who have difficulty writing.

Care in the Process

Key points to consider for a more organized and enjoyable process:



b) First round: placing colored *sticky notes on diagram 3H* (25 minutes)

- Facilitator 1 gives the floor to each participant, in a clockwise direction, so that everyone has the opportunity to express themselves ;

- Each participant reads their aspirations aloud for 1 or 2 minutes ;
- The others do not interrupt; if they disagree, they wait for their turn to speak or the next round, when this will be asked again. Divergences are noted with a numbered asterisk on the sticky notes if described on the flipchart;
- The participant places the sticky note on the board in the upper right corner of the diagram, next to similar aspirations already posted by the other participants;
- Facilitator 1 gives way to the next one, clockwise.

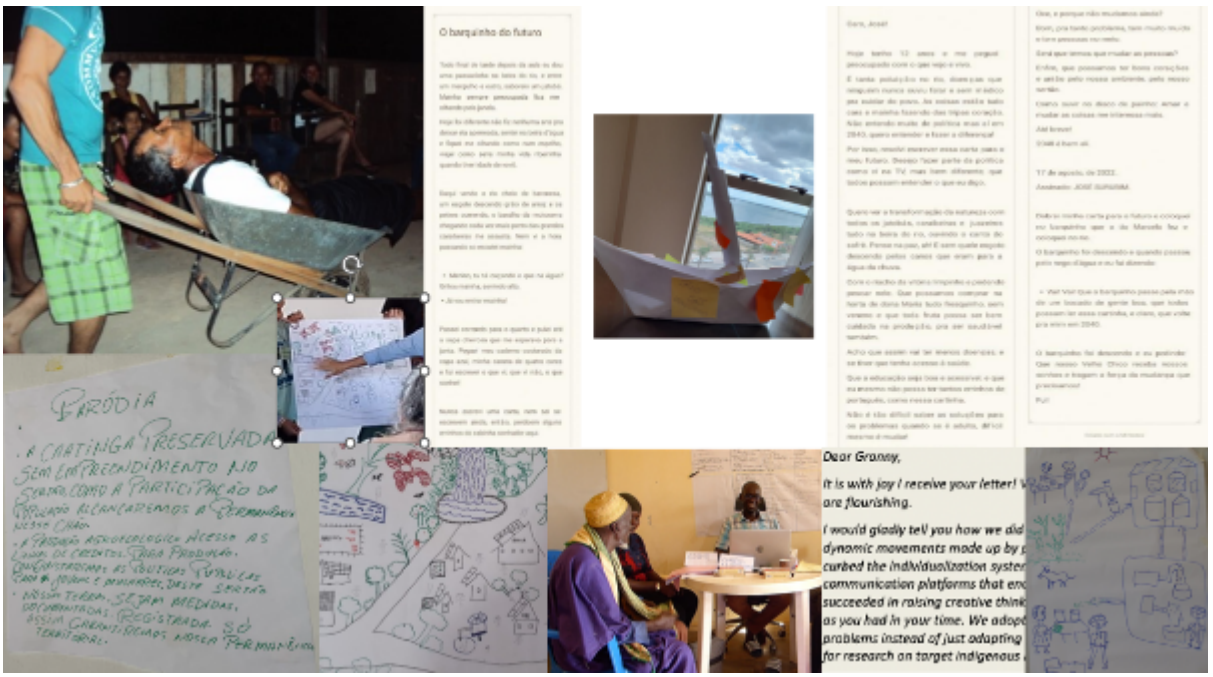
(Meanwhile, facilitator 2 organizes the sticky notes on the board into groups.)

c) Second round - Placing colored sticky notes on diagram 3H (20 minutes)

- Facilitator 1 asks if any other divergences have arisen regarding the aspirations of the other participants. If so, he asks people to explain, and these are noted with an asterisk on the sticky note and flipchart ;
- Ask participants to add new aspirations, of any dimension, that may have emerged after this initial exchange. Check whether any dimension is underrepresented, and encourage participants to add it.
- It also asks them to identify "good seeds" of that desired future that already exists in the present: "Are there projects, initiatives and/or actions that could be a starting point for achieving these aspirations?"

Step 1.2 - Creative Process (30-40 minutes)

In each group, aspirations are synthesized into creative products (letter, narrative, newspaper headlines, tweets, music, podcast, theater, poetry, diary, film script, etc.) that aim, beyond synthesis, at appropriating the dreamed-of future. Each group should be free to choose the format and content and produce its synthesis without the interference (or even the presence) of the facilitators. The creative product is also an effective way to communicate the group's vision of the future to the other participants (and, later, to the general public).



a) Explanation (10 minutes)

- Facilitator 1 explains to the participants that the objective of this activity is a moment of joint reflection, transforming what was discussed into a creative product that synthesizes the stick notes;
- Facilitator 2 distributes examples, such as a letter to a friend or relative dated 2040. Other forms of creative production can be used: music, a play, a newspaper article, poetry, and others. Participants are free to choose the form they prefer.
- Facilitator 1 motivates participants to choose a creative approach and decide who will present it in plenary. They ask whether participants have any questions, then let them work independently.

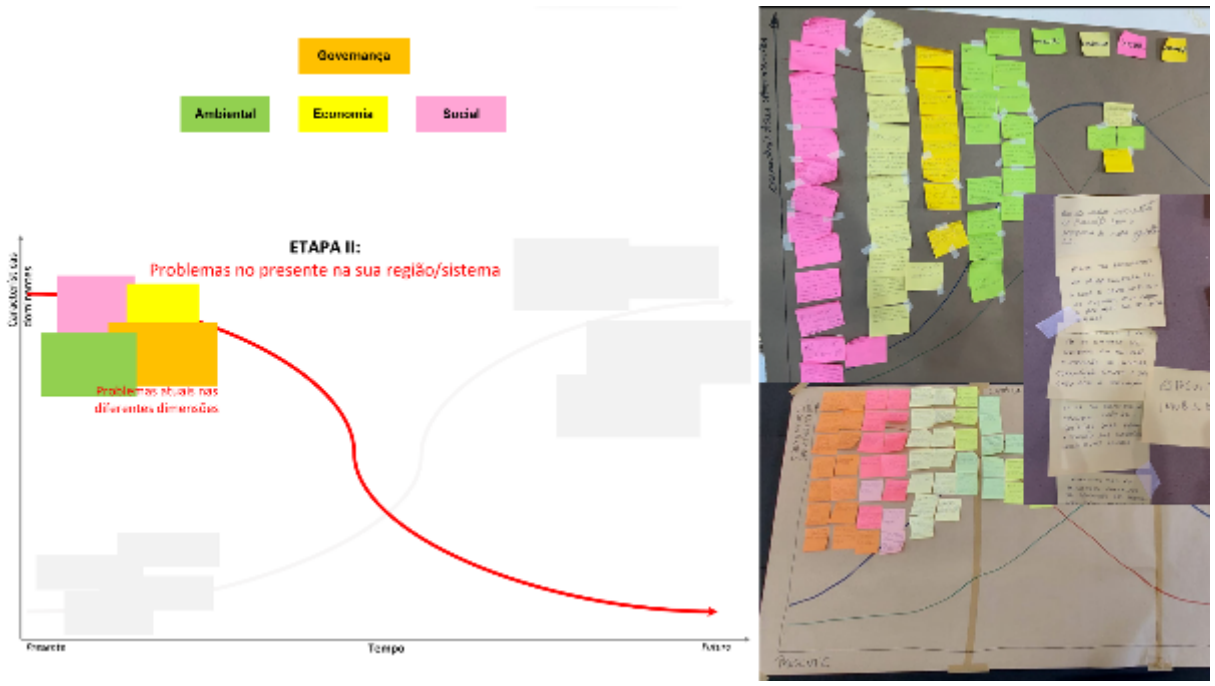
b) Execution (25-30 minutes).

- The facilitators leave the room and emphasize that participants need to prepare their presentation for the other groups, which is usually held right after lunch, before Stage 2 begins.
- The presentations are generally quick (about 5 minutes) but offer an effective way to convey the group's perspectives.

Step 2: Current problems and their causes

Step 2.1 - Present problems in the 3H diagram (45 min)

The process is very similar to Stage 1, but faster. Use the same sticky-note colors, but focus on current problems.



a) Instructions and initial completion of the sticky notes (5 min)

- Facilitator 1 explains the rules of the game:
    - He explains that we will begin with what the participants consider to be the main problems of the present ;
    - Indicate on the 3H diagram, on the table, the location where we will work: upper left corner (H1), red line (Figure 1) ;
    - As in the previous step: review the agreed-upon colors for the sticky notes, for example, representing four dimensions: social (pink), environmental (green), economic (yellow), and governance (orange); explain that consensus is not necessarily the goal; indicate the flipchart *for* any divergences; and inform participants that they should write as individuals, not as representatives of the organization they work for.

- Facilitator 1 asks participants to write on the *sticky notes* :
    - As in the previous step, participants are asked to fill in up to two different colored sticky notes with keywords or short phrases about current problems or challenges, addressing the 4 dimensions; participants are asked to pay attention to their handwriting, preferably making it easy to read; participants are left in silence for 5 minutes to fill in the sticky notes (they usually ask for a few more minutes). Meanwhile, facilitator 2 ensures that everyone has *sticky notes* and pens and helps participants who have difficulty writing.

b) First round: placing colored sticky notes on the 3H diagram (25 minutes)

- As in step 1: Facilitator 1 gives the floor to each participant (for example, clockwise), so that everyone has the opportunity to express themselves; each participant reads their aspirations aloud in 1 or 2 minutes. The other participants are instructed *not to interrupt*; if they have any divergences, they wait for their turn to speak or the next round, when this will be asked. Divergences are noted with a numbered asterisk on the sticky notes and described on the flipchart; the participant sticks the *sticky note* on the board in the upper right corner of the diagram, next to similar aspirations already stuck by the other participants; facilitator 1 gives the turn to the next participant, clockwise. (Meanwhile, facilitator 2 organizes the sticky notes on the board into groups).

c) Second round - Placing colored sticky notes on diagram 3H (15 minutes)

- Facilitator 1 asks if any other divergences have arisen regarding the aspirations of the other participants. If so, he asks people to explain, and these are noted with an asterisk on the sticky note and flipchart ;
- Ask participants to add new problems, of any size, that may have arisen after this initial exchange. Check if any dimension is underrepresented and encourage participants.

Step 2.2 Construction of the CLD(s)

a) Presentation of what a Causal Loop Diagram (CLD) is (10 minutes)

- Facilitator 1 presents slides that explain, in a simplified way, what a CLD is, its purpose, and why it matters in this context. Box 4 summarizes some instructions, provides video links, and the slides we use. In general, each group builds its own CLD and then presents it to the other group.

b) Defining central themes (15 minutes)

- Next, facilitator 1 asks participants to collectively prioritize **three. Problems** are chosen based on their importance to the region or system being transformed. *The* selected problems are typically seen as the tip of the iceberg ;
- Selection criteria:
    - Priority (urgency);
    - A problem over which participants feel they have agency;
    - Central: it encompasses several aspects.
- If the participants cannot reach a consensus, a vote can be organized. Use the *flipchart* to assist in the selection process.

c) CLD Construction (40 minutes)

- Start the CLD with these three problems, distributing them on a sheet of paper on which the CLD will be drawn. The diagram is usually drawn on paper, using *sticky notes* or a pencil to facilitate moving and modifying flows[6];
- The exercise continues by encouraging participants to reflect on the causes of the variables in the diagram to examine deeper factors. The facilitators' role is to question participants inclusively and help them construct the diagram.

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[6]Software like Vensim can also be used, although in our experience, using paper—with pencils being shared by participants—is more inclusive. Software createsParticipants then add new variables and causal relationships to the CLDs as they distance.

- Participants then add new variables and causal relationships to the CLDs as they iteratively identify them. Topics previously listed as problems in exercise 3H or new ones may be included. In many cases, from a certain point, the participants themselves begin to assemble the diagram independently ;
- Whenever possible, seek causal circular relationships, also known as feedback loops (see Box 2 and Box 4). But the most important thing is to capture the relationships and understand the causes from the participants' perspective. Very important: if there are divergences, as in the previous steps, mark them with red asterisks and note them separately. When in doubt, also indicate this with a dashed line ;
- The exercise typically lasts forty to fifty minutes in groups of ten to twelve participants.

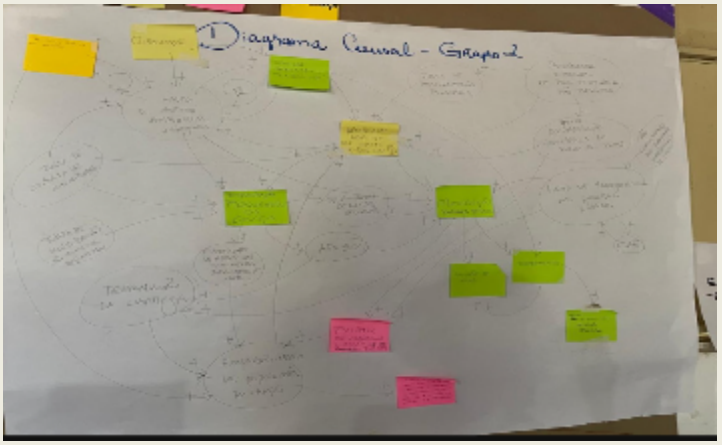
Box 4 - Guidelines for the Collective Construction of CLDs

The Causal Loop Diagram (CLD) is a visual tool for understanding the root causes of problems. The goal is not to create a "perfect" model, but to capture the group's systemic and collective perception.

Materials:

Large white paper, *sticky notes* , pencil and eraser (make changes easier).

Thick pens for visibility and red markers for discrepancies.



Practical Steps for Construction:

List: Identify the key variables (use nouns that can "increase" or "decrease"; avoid verbs).

Connect: For each pair, ask: "if A increases, what happens to B?" and define the polarity (+ or –).

Plot: Draw the arrows connecting the variables.

Closure: Identify loops and classify them as Reinforcing (R) or Balancing (B), marking them in the center of the loop.

Delays: If an effect is delayed or takes time to manifest, indicate this with two slashes (‖) above the arrow (delays generate oscillations and counterintuitive behaviors).

Cycles : Always encourage circular (cyclical) systemic thinking, rather than linear chains.

Facilitation Principles:

Facilitator's role: Guide the process without doing the work for the participants. Encourage them to express their understanding using their own language and foster autonomy (an indicator of success is when they begin to conduct it on their own).

Inclusion: Ensure everyone participates and contributes. If a few people monopolize the discussion, summarize the process in rounds.

Markings on the Diagram:

Discrepancies: Indicate with red asterisks (*) on the stick notes if noted separately on the flipchart.

Questions: Use dashed arrows to indicate causal relationships that require further or future discussion.

Contingency Tips:

Large groups (>12 people): Separate into subgroups and integrate the results/diagrams later.

Too many variables: Prioritize the central variables (those with the greatest impact on the system, which, if modified, would affect the largest number of others) and record the remaining variables separately on the *flipchart* for later analysis.

Limited time: Focus and prioritize identifying the main loops instead of exhaustively detailing each link.

Preparation for the Post-Workshop:

Photograph the diagrams and transfer them to Vensim as soon as possible (ideally before STEP 3).

Adjustments or new interpretations should be transparent and subject to review and validation by the participants.

Step 3: Necessary actions

What strategies and actions are needed to identify the main problems and their respective causes? What actions are needed to "grow the good seeds"?

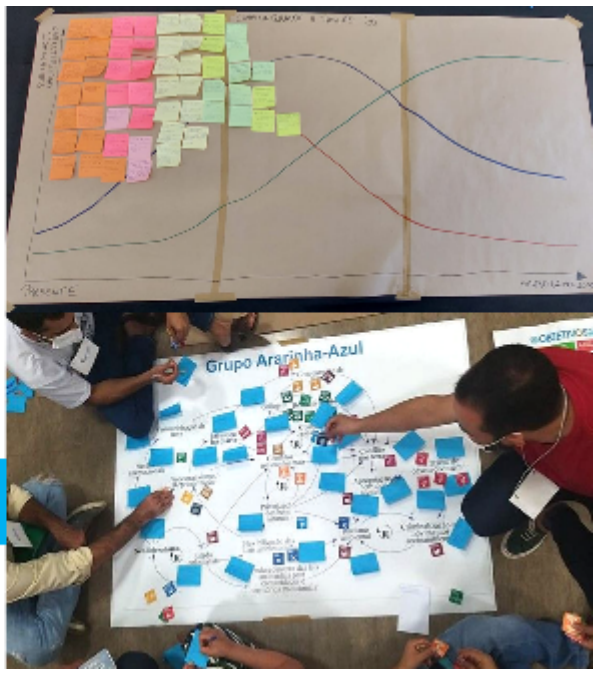
This step is performed based on the CLDs (Causal Loop Diagrams). The focus is on identifying, within the systemic structures, the points to be modified and the necessary actions.



CLDs

Problemas

Ações



Step 3.1 - Leverage points and actions in CLDs (40 minutes)

- Facilitator 1 asks people to circulate around the room (analyzing the CLDs and the good seeds), discuss with others, and write (together) on blue sticky notes.

  What needs to be done and by whom? (actors with the power to intervene in that strategy/action) Is it possible to prioritize actions in the short, medium, and long term?

- Facilitator 1 asks everyone to read, and Facilitator 2 pastes it next to the problem.
- Facilitator 1 asks if they want to add anything.

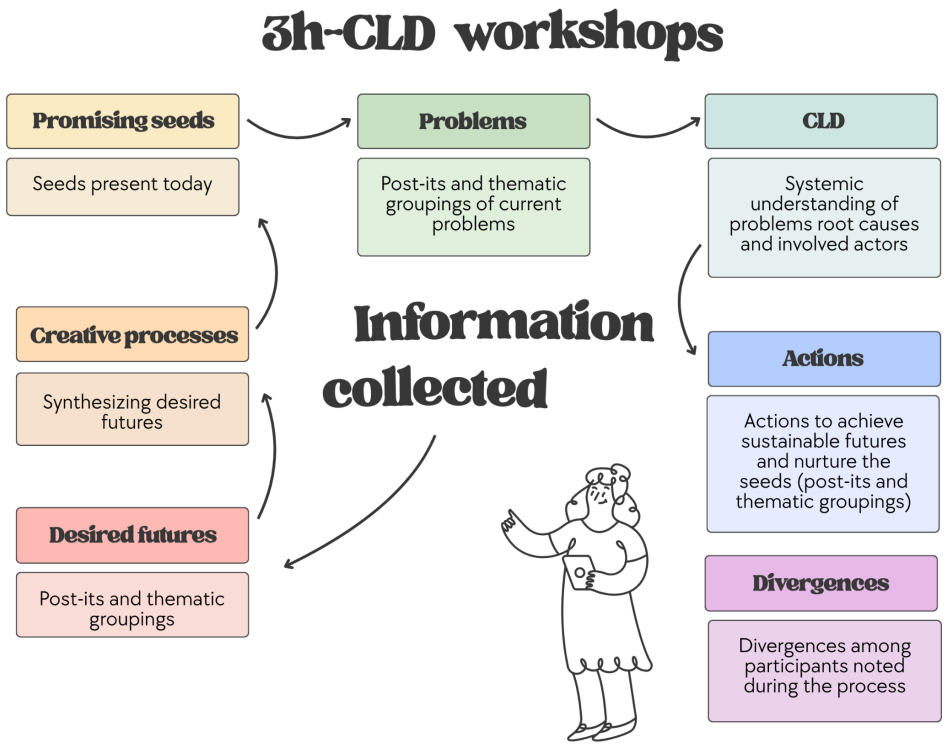
Step 3.2 Presentation of results (30 min)

One group presents the results to the other. And, after that, the facilitators guide the workshop to its conclusion.

Other tips about the workshops

A dialogue process based on the 3H-CLD method can be organized in a single workshop or multiple workshops, depending on the objective. Each workshop can address a scale or a stage of the process, for example. In multiscale and/or comparative studies between case studies (such as the NEXUS-XPaths and Bridging projects, for example), STEPS 1, 2, and 3 are repeated several times with different actors – and only then, in the Post-Workshop activities, are the results integrated/compared. Regardless of the configuration, at the end of the process, the following information will be available for each workshop/group:

1. Desired Futures (*stick notes* if grouped by similar themes);
2. Creative processes synthesizing desired futures;
3. Good seeds already exist today;
4. Current problems (*stick notes* grouped by similar themes);
5. Systemic understanding of the root causes of the problems and the actors involved (CLD);
6. Actions to achieve sustainable futures and grow the seeds (stick notes grouped by similar themes);
7. divergences between participants.



IMPORTANT: At the end of each workshop, all materials produced by the groups must be photographed, transcribed, and organized. This includes sticky notes, CLDs (Creative Learning Diagrams), proposed actions, results of creative processes, notes from facilitators and co-facilitators; divergences, questions, and observations recorded during the activities should also be preserved. The organization should allow for the identification of the origin of each contribution, maintaining the relationship between the group, the workshop stage, and the original material.

POST-WORKSHOP: ORGANIZING AND ANALYZING THE DATA

Transcription, organization and validation

This stage involves preparing the materials for subsequent analysis. The transcription and organization stage consists of:

1. **Literal transcription of the *stick notes*:** Transcribe the *stick notes* by group exactly as they were written by the participants, correcting only any spelling/grammar errors.
2. **Grouping the *stick notes*:** Group the *stick notes* by theme, without needing to distinguish between dimensions [7]. This thematic analysis can be started during the workshops, if there is time, but reviewed after the workshop and validated with the participants. Theme names should correspond to factors associated with each stage. For example, "lack of technical assistance" in STAGE 2 (problem) or "technical assistance" in STAGE 1 (aspiration). Count the number of sticky notes in each theme (important for later analyzing more important factors, comparing case studies, etc.). The appendices to the NEXUS project reports (https://zenodo.org/records/10197094) provide examples.
3. **Review of CLDs :** Digitize the CLDs (in software such as Vensim) without significantly altering the variables and connections built by the participants. Optionally, create integrative CLDs that reflect a synthesis of the CLDs generated in the workshop(s) by different groups. *Validate any changes, improvements, corrections, and integrative CLDs with the participants.*
4. **Validation**: Once transcribed and organized, send the material to participants for validation. Any adjustments, new interpretations, or causal relationships suggested later should be transparently recorded (e.g., marked in red) and submitted to participants for review.

[7] In the case of sustainability, these were used only to ensure they weren't forgotten during the workshop – but the goal is to address the problems and aspirations described in a holistic way.

More details on the data organization and validation stage are provided in Aguiar et al. (2023) and Aguiar et al. (2025).

Co-creating Strategic Action Plans

After the transcription, organization, and validation of the materials, the process can advance to the co-construction of one (or more) Strategic Action Plan(s). In this case, the objective is to translate the knowledge produced collectively into concrete paths to achieve the sustainable and just futures envisioned by the participants.

The plan should not be a list of isolated solutions, but rather reflect an understanding of the relationships between the problems, their root causes, and the structures that maintain the system in its current state.

The plan's development stems from the articulation between the three moments of the 3H-CLD: desired futures and existing transformative initiatives; current problems, their causes, and systemic relationships; and the actions necessary to promote transformation. CLDs help identify key variables, feedback loops, and leverage points where an intervention can produce broader, more significant change.[8] Based on this analysis, participants can prioritize the problems that need to be addressed and define strategic actions with the greatest transformative potential.

Experience with the NEXUS-XPaths project shows that this plan can be the result of a continuous process of dialogue, analysis, synthesis, and validation. After a series of multi-scale participatory workshops, the materials produced were analyzed and returned to the participants. In a new workshop, participants jointly analyzed a synthesis of the CLDs (Common Landmarks) and the actions proposed during the workshops, selected central and interrelated problems, and detailed strategic actions for each. In this process, the SDGs (Sustainable Development Goals) were used as a broad reference for sustainability, without replacing the

[8]In projects where multiple workshops were held, it may be necessary to integrate the information from the CLDs, creating integrating CLDs . The process is participants jointly analyzed a synthesis of the CLDs (Common Landmarks) and the actions proposed during the workshops, selected central and interrelated problems, and detailed strategic actions for eachdescribed in detail in (Aguiar et al. , 2023, 2025) NEXUS-XPaths.

priorities, experiences, and needs identified in the territory itself (Sonetti -González et al., 2026). Each strategic action may present:

- the problem and the systemic causes that you intend to address;
- the desired transformation and its objectives;
- the leverage point where you seek to intervene;
- the activities necessary for its implementation;
- The actors involved and their responsibilities;
- the territories, social groups and scales considered;
- the necessary institutional resources and conditions;
- the implementation horizon;
- potential obstacles, risks, conflicts, and unintended consequences;
- the synergies and tensions with other actions;
- methods of monitoring, learning and assessment;
- Its relationship with public policies, existing initiatives, and the SDGs.



The Action Plan should maintain the link between each proposal and the results built during the participatory process. It should also indicate who can benefit from the actions, who can be affected, and what power asymmetries need to be considered. Convergences, divergences, and open questions should remain visible and be submitted for collective validation. In this way, the plan becomes a living document that can be revised as new experiences, knowledge, and conditions emerge. The table below presents a simplified example of a form used to build strategic action plans in the Bridging Project. Values.

Action Plan

Name of Strategic Action: Program, etc.				
Description: [an action targeting a CLD "leverage point"]				
Rationale: [how it will solve the problem]				
Those responsible for execution: [who among the participant group will coordinate the action's implementation]				
Actors involved/roles: [which other actors are involved in the action's implementation]				
Obstacles: [main difficulties, opposing actors, etc.]				
Opportunities: [e.g., ongoing policies]				
Implementation	Stages	Tasks (optional)	Timeframe (optional)	Resources (optional)

For examples of action plan development processes, including in other countries, see the XPaths project results. (López-Rodríguez, Jiménez-Aceituno and Castro Martínez, 2024; Sonetti González et al. , 2024; Diallo et al. , 2026) and the BridgingValues Brasil project (https://www.ccst.inpe.br/bridgingvalues/).

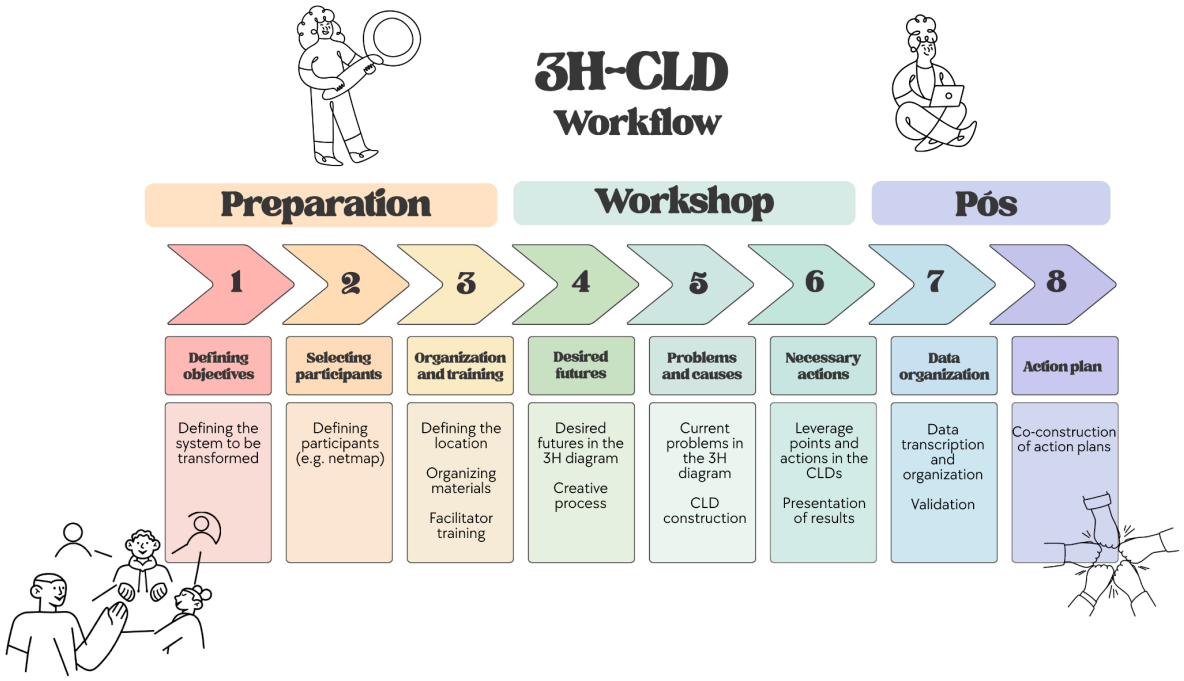
Other analyses can be performed using the workshop results. Among them, we highlight the construction of narratives for target-seeking scenarios representing pathways to sustainable and just futures. The alternative narratives derive from the divergences (between participants, groups, and scales) identified during the 3H-CLD workshops, following the approach detailed in (Aguiar *et al.*, 2020). This process was applied in the NEXUS project (https://nexus.ccst.inpe.br/), and detailed in the scenario publications (Aguiar et al., 2020, 2026). The international projects XPaths (https://www.xpathsfutures.org/) and Bridging Values (https://www.bridgingvalues.ch/) carried out a series of comparative analyses across countries and scales (López-Rodríguez et al., 2024; XPaths Project Team, 2024).

FINAL CONSIDERATIONS

This document presented a guide for applying the 3H-CLD method, and here we reinforce the importance of careful execution of the three phases:

- **Preparing workshops as a foundation:** Without precise stakeholder mapping and a clear definition of objectives, the workshop phase loses its power because it lacks the voices fundamental to systemic change.
- **Workshops as a driving force:** This is when we use the tools of the Three Horizons model and causal diagrams to discuss the desired future, the problems of the present, and the actions needed to transform the system.
- **Post-Workshop as support for action:** Rigor at this stage ensures that the knowledge generated is not lost, transforming the synthesis into co-constructed strategic action plans or other analyses.

We hope it will be useful as a tool for social transformation.



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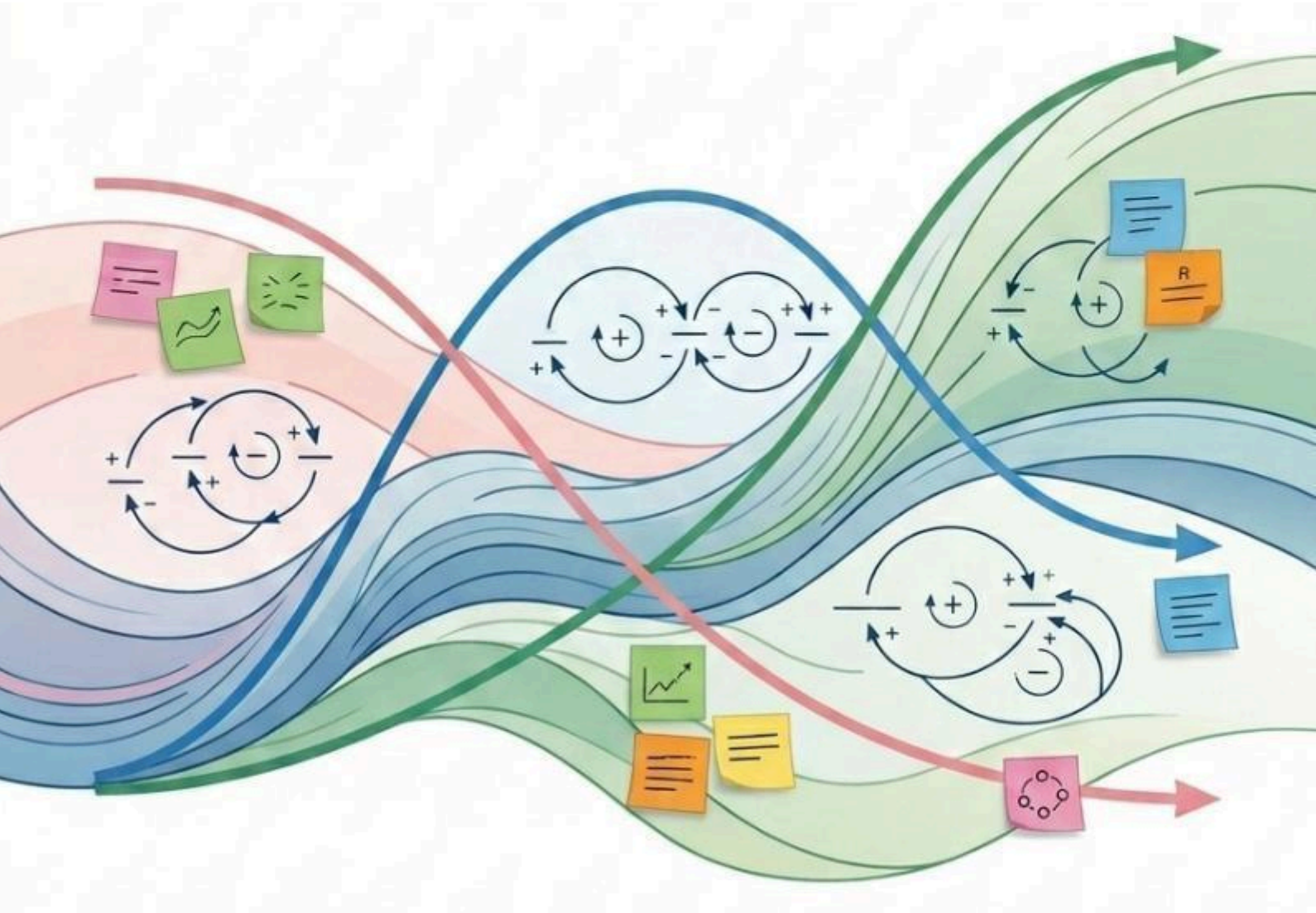
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REALIZATION